

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA0304

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO2: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Dr DUMPALA PRASANTH

Code of Conduct:

I (V Sunil Kumar/ 192425292) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

V Sunil Kumar

Signature of the student:

QUESTIONS: 10

Total Marks: 50

Annexure A
Application - Based Questions

Duration: 2Hrs

1. Design and develop a Reinforcement Learning framework a ride-sharing company uses Reinforcement Learning to match drivers with passengers efficiently. Apply RL concepts to define the state space, action space, and reward function to minimize waiting time and improve service quality. **(5 Marks)**

ANSWER:

1. Understanding of the Problem

A ride-sharing company wants to match drivers with passengers quickly to reduce waiting time and improve customer satisfaction.

2. Application of RL Concepts

State Space

- Driver location
- Passenger location
- Traffic condition
- Available drivers

Action Space

- Assign nearest driver
- Reassign another driver
- Wait for a better match

Reward Function

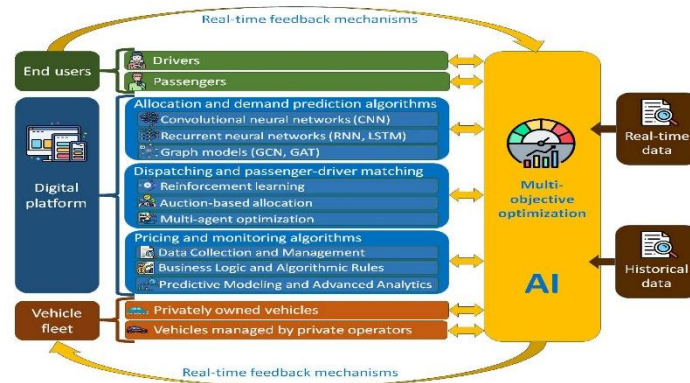
- Positive reward for quick pickup and successful ride
- Negative reward for delays and cancellations
-

3. Problem-Solving Approach

The RL agent observes the current state, selects the best driver, receives rewards based on service quality, and updates its policy through continuous learning.

4. Solution/Outcome

- Reduced waiting time
- Better driver utilization
- Improved customer satisfaction



2. An e-learning platform uses Reinforcement Learning to recommend personalized study materials. Recall how exploration and exploitation strategies help balance familiar and new content recommendations. **(5 Marks)**

ANSWER:

1. Understanding of the Problem

A ride-sharing company wants to match drivers with passengers quickly to reduce waiting time and improve customer satisfaction.

2. Application of RL Concepts

State Space

Driver location

Passenger location

Traffic condition

Available drivers

Action Space

Assign nearest driver

Reassign another driver

Wait for a better match

Reward Function

Positive reward for quick pickup and successful ride

Negative reward for delays and cancellations

3. Problem-Solving Approach

The RL agent observes the current state, selects the best driver, receives rewards based on service quality, and updates its policy through continuous learning.

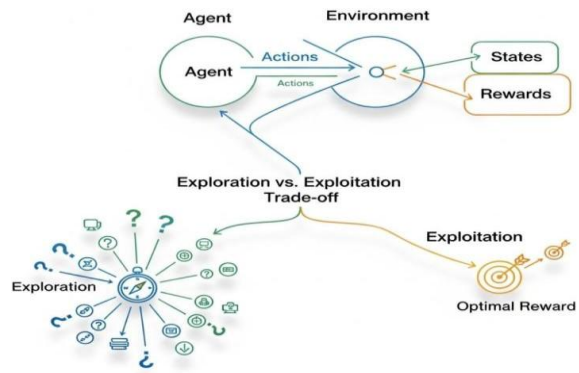
4. Solution/Outcome

Reduced waiting time

Better driver utilization

Improved customer satisfaction

How Reinforcement Learning Works



3. Design and develop a Reinforcement Learning framework a robotic vacuum cleaner uses Reinforcement Learning to clean efficiently while avoiding obstacles. Apply the concept of policy and explain how the value function improves performance over time.

(5 Marks)

ANSWER:

1. Understanding of the Problem)

The robot must clean efficiently while avoiding obstacles.

2. Application of RL Concepts

Policy

- Selects the next movement
- Avoids obstacles

Value Function

- Estimates future rewards
- Chooses efficient cleaning paths

3. Problem-Solving Approach

The robot learns from experience by receiving rewards for efficient cleaning and penalties for collisions.

4. Solution/Outcome

- Faster cleaning
- Better obstacle avoidance
- Reduced battery usage



4. A cryptocurrency trading system uses Reinforcement Learning to decide buy, sell, or hold actions. Outline the challenges in designing reward functions and explain how poor reward design can affect performance. **(5 Marks)**

ANSWER:

1. Understanding of the Problem

The trading system decides when to buy, sell, or hold cryptocurrencies.

2. Application of RL Concepts

Reward Challenges

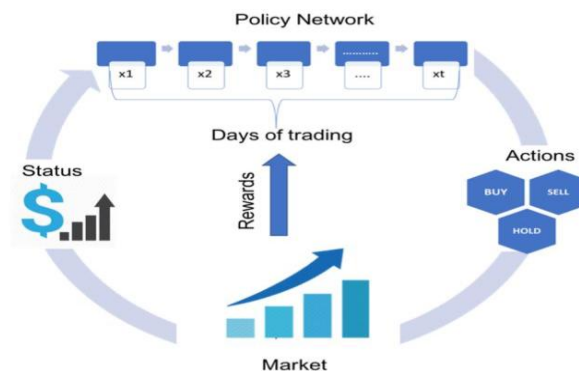
- Price volatility
- Delayed profits
- Market uncertainty

3. Problem-Solving Approach

The RL agent updates its trading strategy using rewards based on long-term profit.

4. Solution/Outcome

- Better trading decisions
- Reduced financial loss
- Improved profitability



5. Develop a smart grid system uses Reinforcement Learning to manage electricity distribution. Create an RL model by defining state space, action space, and reward mechanism.

(5 Marks)

ANSWER:

1. Understanding of the Problem

The smart grid manages electricity distribution efficiently.

2. Application of RL Concepts

State Space

- Electricity demand
- Power generation
- Battery level

Action Space

- Increase supply
- Reduce supply
- Store energy

Reward Function

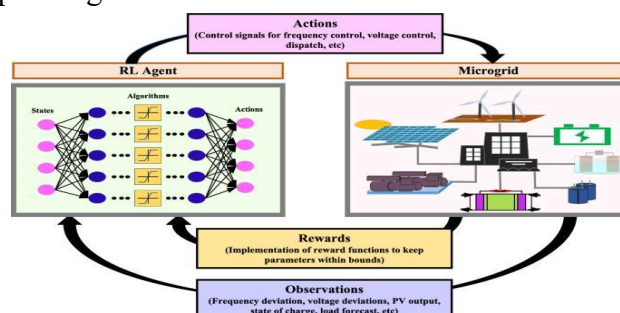
- Positive reward for balanced distribution
- Negative reward for power loss

3. Problem-Solving Approach

The RL agent continuously monitors demand and adjusts power distribution.

4. Solution/Outcome

- Reliable electricity supply
- Lower energy wastage
- Reduced operating cost



6. A healthcare monitoring system uses Reinforcement Learning to adjust medication dosages. Apply RL concepts to define states, actions, and rewards, and explain how learning improves outcomes.

(5 Marks)

ANSWER:

1. Understanding of the Problem

The system adjusts medication dosage according to the patient's health.

2. Application of RL Concepts

States

- Patient condition
- Blood pressure
- Heart rate

Actions

- Increase dosage
- Reduce dosage
- Maintain dosage

Rewards

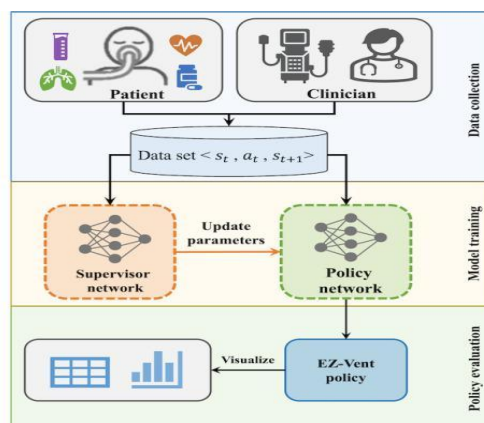
- Positive reward for health improvement
- Negative reward for side effects

3. Problem-Solving Approach

The RL agent learns the optimal dosage by analyzing patient responses.

4. Solution/Outcome

- Personalized treatment
- Better recovery
- Fewer health risks



7. Design a model drone delivery system uses Reinforcement Learning for navigation. Recall how exploration and exploitation strategies influence route optimization.

(5 Marks)

ANSWER:

1. Understanding of the Problem

The drone must deliver packages using the best route.

2. Application of RL Concepts

Exploration

- Tries new routes

Exploitation

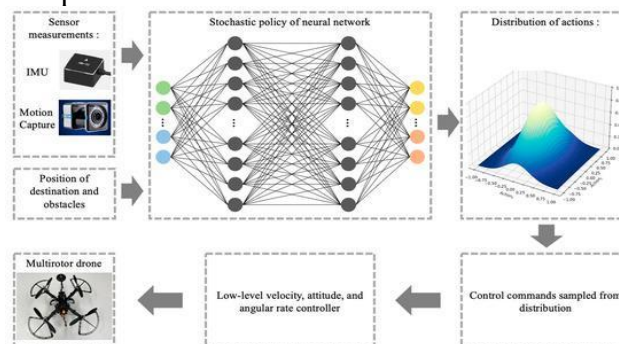
- Uses successful routes

3. Problem-Solving Approach

The RL agent balances exploration and exploitation to find optimal routes.

4. Solution/Outcome

- Faster deliveries
- Reduced battery consumption
- Better route optimization



8. **Design and develop** a Reinforcement Learning framework a manufacturing robot uses Reinforcement Learning to optimize assembly operations. Apply the concept of policy and explain how value functions help improve efficiency.

(5 Marks)

ANSWER:

1. Understanding of the Problem

The robot performs assembly operations efficiently.

2. Application of RL Concepts

Policy

- Chooses assembly actions

Value Function

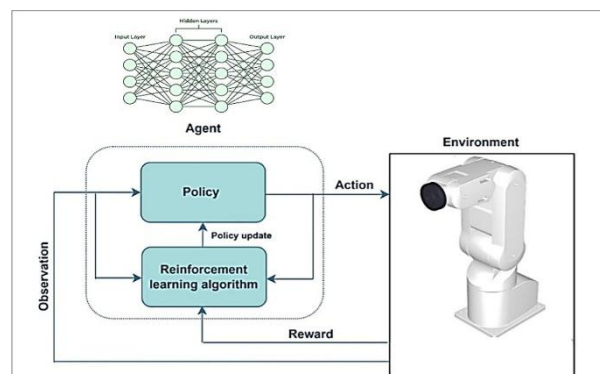
- Estimates long-term rewards

3. Problem-Solving Approach

The RL agent improves its policy based on rewards from successful assembly operations.

4. Solution/Outcome

- Faster production
- Fewer errors
- Higher product quality



9. An online advertising system uses Reinforcement Learning to maximize click-through rates. Outline reward design challenges and explain their impact on system performance.
(5 Marks)

ANSWER:

1. Understanding of the Problem

The system aims to maximize advertisement click-through rates.

2. Application of RL Concepts

Reward Challenges

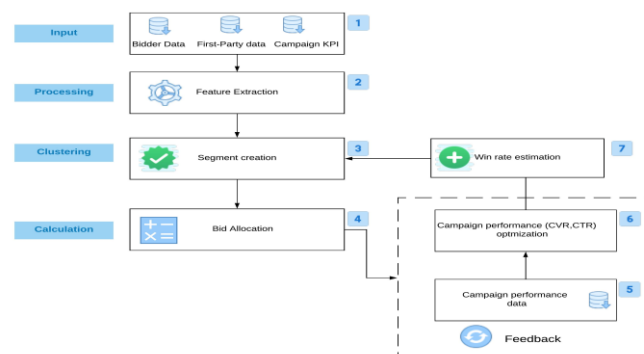
- Delayed clicks
- Fake clicks
- Changing user preferences

3. Problem-Solving Approach

The RL agent updates advertisement selection using user interaction rewards.

4. Solution/Outcome

- Higher click-through rate
- Better ad targeting
- Increased revenue



10. A smart irrigation system uses Reinforcement Learning to optimize water usage. Create an RL framework by defining states, actions, and reward function. **(5 Marks)**

ANSWER:

1. Understanding of the Problem

The irrigation system supplies the right amount of water to crops.

2. Application of RL Concepts

State Space

- Soil moisture
- Weather
- Temperature

Action Space

- Start irrigation
- Stop irrigation
- Adjust water flow

Reward Function

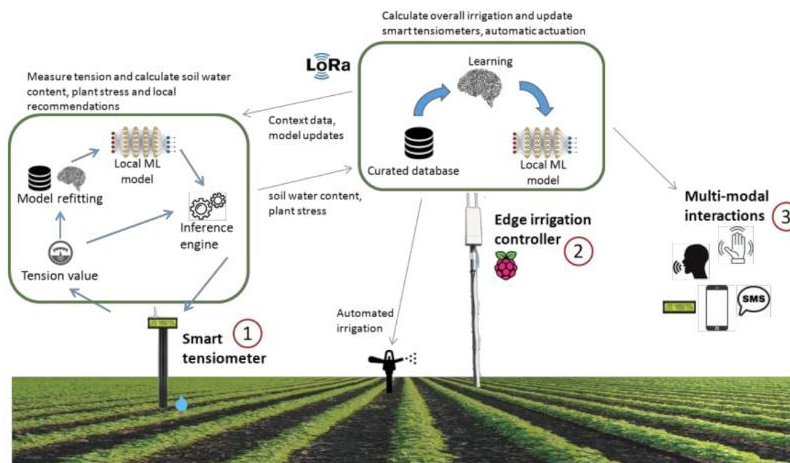
- Positive reward for saving water
- Negative reward for overwatering

3. Problem-Solving Approach

The RL agent learns the best irrigation schedule from environmental feedback.

4. Solution/Outcome

- Efficient water usage
- Better crop growth
- Sustainable farming



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand